Kairav Session 28

Sunday, September 27, 2026 8:33 PM

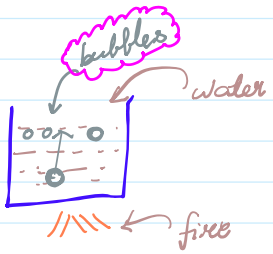
Sorting :- It is the process of arranging element increasing or decreasing fashion.

Sorting Algorithm :-

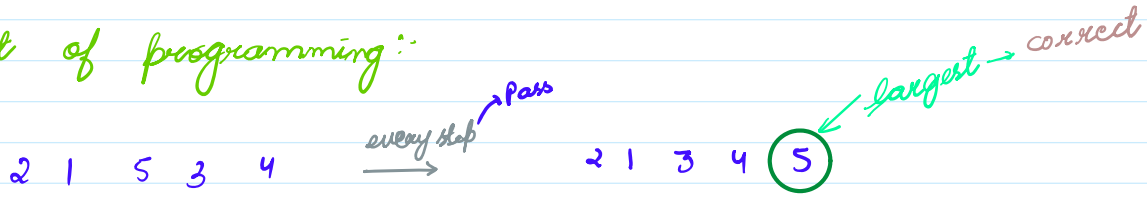
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Counting Sort
5. Merge Sort
6. Quick Sort

Bubble Sort :-

Real life Example :-



Content of programming :-



-> In bubble sort at every pass the *largest element get bubbled up* (will move towards the end of the list)

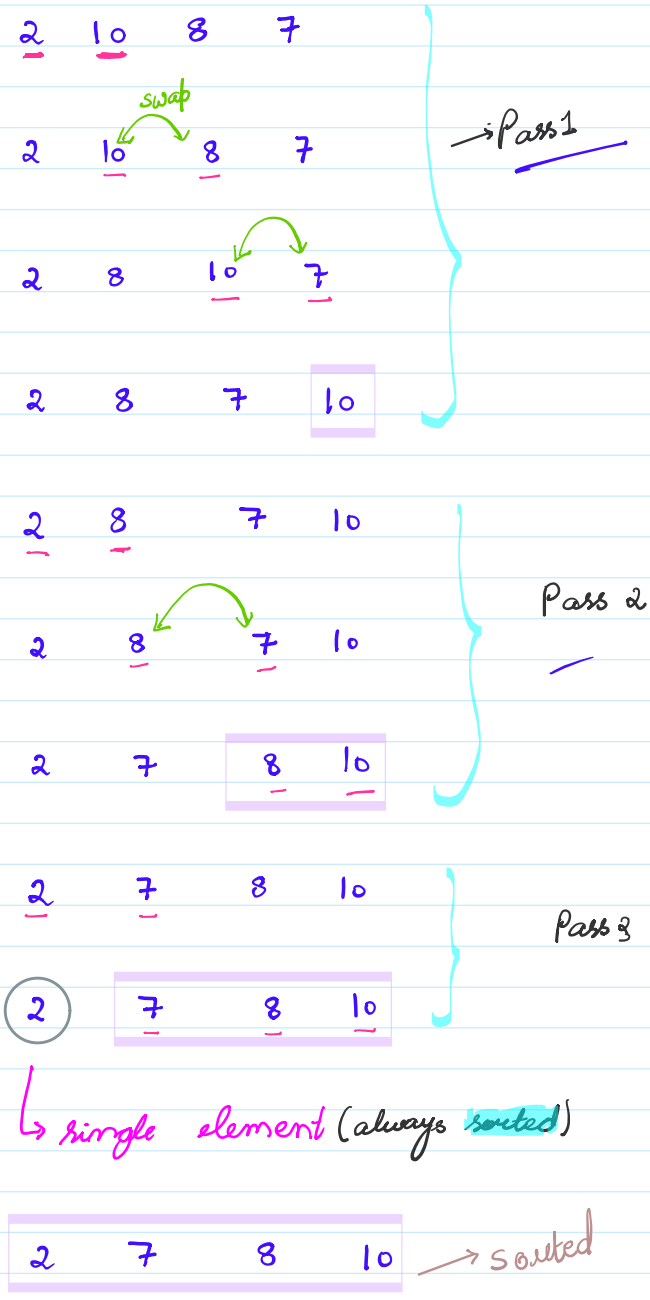
Algorithm :-

→ It keeps comparing the elements if they are in wrong order we will exchange its position (swapping)

Swapping :-

x = 10 y = 5 —swapping→ x = 5 y = 10

Step by Step :-



↳ single element (always sorted)

2 7 8 10 → sorted

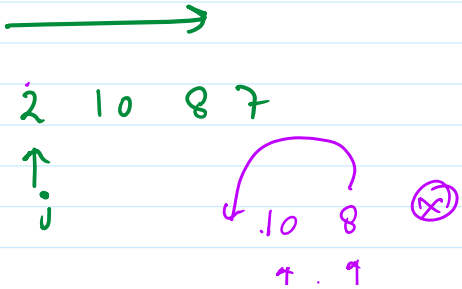
Pseudo Code :-

$n = 4$ Pass = 3

Passes = $n-1$

$n-1$

i = passes



```
for i in range(Pass):          # Pass = n-1
    for j in range(n-1-i):
        if (arr[j] > arr[j+1]):   # wrong order
            swap(arr, j, j+1)
```

i

4 10 8 (x)

j j+1